are likely to allow uncontroversial evaluation (by priming for short model outputs), though later we also consider open-ended generation (Section 7). In addition, a subset of the test suite considers idioms, which have both literal and figurative interpretations, as concepts in the test prompts.

We manually create 109 prompts with concepts from categories such as colors, food, animals, songs, occupations and more. A few examples are listed in Table 1.[2] Each prompt in our test suite is matched with a control prompt as explained above. When designing the control prompts, we prioritized keeping them as basic and simple as possible to avoid confounding factors. In some cases, more elaborate prompts required us to include richer control prompts that are more similar to the test prompts in their length and structure.[3] Both types of control prompts are included in our evaluation.

3 Experimental Setup

We present the experimental setup for automatically evaluating semantic leakage in a wide range of models and model sizes.

3.1 Experimental Details

We evaluate semantic leakage in multiple language models from two families: GPT[4] and LLAMA models (Touvron et al., 2023), as detailed below. For all models, we explore several temperature values (0, 0.5, 1, 1.5), and run each prompt 10 times to get variation in the generations, when possible.

Before evaluation, for cases where the prompt is repeated in the model generation, we remove the repeated prompt. We also truncate the generations after the first period since the main piece of information is generated before it, and because LLAMA models tend to generate unrelated sentences or phrases that might interfere with the evaluation.

3.2 Models

We experiment with 13 models of two different state-of-the-art model families to explore semantic leakage in a diverse inventory of models.

GPT models We use OpenAI's API and send requests to GPT models by calling GPT3.5, GPT4 (Achiam et al., 2023), and GPT4O.[5] With GPT models, for sentence completion prompts we prepend "Complete the sentence:", as we find the model performs the task better this way.

LLAMA models We run all LLAMA variations using Huggingface (Wolf et al., 2019).[6] We cap the generation in LLAMA models at 100 tokens (300 tokens for open generation, Section 7).

3.3 Embedding Methods

For automatic evaluation, we aim to use basic embedding methods that are able to detect and reflect semantic similarities, and are ideally detached from the models we evaluate to avoid confounding factors. We consider the following embedding methods. BERT-SCORE directly provides a similarity score, for the others we apply cosine-similarity.

BERT-SCORE (BS) BERT-SCORE (Zhang et al., 2020) is an automatic evaluation metric for text generation, that computes a similarity score for each token in the candidate sentence with each token in the reference sentence, where token similarity is computed using contextual embeddings. We use the `distilbert-base-uncased` model.[7]

For the multilingual experiments, where we expect generations in non-English languages (Section 6), we use the respective models: `bert-base-chinese` for Chinese, and `bert-base-multilingual-cased` for Hebrew. Crosslingual settings still use the English model as the generations there are mainly in English.

SENTENCEBERT EMBEDDINGS (SB) SENTENCEBERT (Reimers and Gurevych, 2019) is a modification of BERT (Devlin et al., 2019) that uses Siamese and triplet network structures to derive semantically meaningful sentence embeddings that can be compared using cosine-similarity. We use the huggingface implementation.[8]

OPENAI EMBEDDINGS (OAI) We use `text-embedding-3-large`,[9] OpenAI's best

[2] The full list can be found at https://gonenhila.github.io/files/semantic_leakage_prompts.txt

[3] An example is the prompt *"She went on a field trip to the zoo. Her sister got a gift from a friend: a shirt with a print of"* for which we design the control prompt: *"She went on a field trip. Her sister got a gift from a friend: a shirt with a print of"*.

[4] https://platform.openai.com/docs/models

[5] `gpt-3.5-turbo-0125`, `gpt-4-turbo-2024-04-09` and `gpt-4o-2024-05-13`, respectively.

[6] LLAMA2: 7B, 7B-chat, 13B, 13B-chat, 70B, 70B-chat. LLAMA3: 8B, 8B-Instruct, 70B, 70B-Instruct.

[7] https://huggingface.co/spaces/evaluate-metric/bertscore

[8] https://huggingface.co/efederici/sentence-bert-base

[9] https://platform.openai.com/docs/guides/embeddings/embedding-models

Model	Prompt and generation
LLAMA2 7B-CHAT	*John likes dolphins. John's father is working as a* marine biologist
	John likes tigers. John's father is working as a tiger in a zoo
LLAMA2 13B-CHAT	*He likes pandas. His favorite food is* bamboo
	He likes mice. His favorite food is cheese
LLAMA2 70B-CHAT	*His name is Cedar. His friend lives in* a treehouse
	His name is Ocean. His friend lives in a houseboat
LLAMA3 8B-INS	*He watched a polka concert. For dinner he had* a plate of pierogies and a side of sauerkraut
	He watched a flamenco concert. For dinner he had a delicious paella
LLAMA3 70B-INS	*Mercedes has a friend. The friend's hobby is* collecting Mercedes cars because he loves the Mercedes brand
	Forest has a friend. The friend's hobby is gardening, to her home in a way which makes it a forest
GPT3.5	*He is a firefighter. His favorite song is* "Ring of Fire" by Johnny Cash
	He is a postman. His favorite song is "Signed, Sealed, Delivered" by Stevie Wonder
GPT4	*The woman was as cold as ice. Afterwards, I went to the library and read* about the fascinating science of cryogenics, which explores the process of freezing and preserving organic matter at extremely low temperatures
	It was her turn to speak and she got cold feet. A day later she stayed at home because she got hurt in her ankle during a morning jog
GPT4O	*She is a music lover. Her nephew was accused of* stealing her prized vinyl collection
	She is a cinema lover. Her nephew was accused of pirating movies

Table 1: Examples of instances that lead to semantic leakage, taken from different models. The prompt is in green (*italic*) and the generation is in blue.

performing embeddings. There is no public documentation of the model training.

In addition to these metrics, we validate our experiments with a manual evaluation of semantic leakage on a subset of the models (Section 5). This also serves as a validation of our automatic metrics.

4 Results

Significant semantic leakage across various use cases and models. Table 2 depicts the average leakage for each model, across multiple samplings and temperature values, as detailed in Section 3.1. We see that semantic leakage is exhibited by all model variations, and is detected by all embedding models we use. Leak-Rate values are all well above the 50% random mark and statistically significant[10] with $p < 10^{-100}$, validating the semantic relatedness of the prompt and the test generation.

Table 1 lists a few examples of semantic leakage from the different models, showcasing leakage in diverse use cases and styles and with respect to a variety of leaking concepts. In many cases we explore, the generations do not make sense in the context, or are very limited and focused on the leaking concept from the prompt.

Leakage is more pronounced in certain model variations. The results in Table 2 show that certain model variations tend to exhibit more semantic leakage than their counterparts. We now analyze the differences within the 2 model families.

Model	Leak-Rate BS	SB	OAI
GPT3.5	74.3	68.6	**85.5**
GPT4	70.8	61.2	84.4
GPT4o	**76.9**	**70.4**	85.0
2-7b	66.8	64.9	72.8
2-7b-chat	72.6	**71.7**	77.8
2-13b	70.4	65.1	73.6
2-13b-chat	71.5	65.2	**78.4**
2-70b	71.9	63.2	75.9
2-70b-chat	**75.4**	66.8	78.2
3-8b	69.6	65.9	75.5
3-8b-ins	**78.1**	68.8	**81.5**
3-70b	71.6	68.1	75.2
3-70b-ins	76.3	**71.2**	77.3

Table 2: Semantic Leak-Rate averaged across 10 samples for each of 4 temperature values. No semantic leakage would correspond to a Leak-Rate of 50% (random guessing), with higher values indicating more leakage. The bottom sections present the LLAMA2 and LLAMA3 models, respectively. The model showing the most leakage in each <model family, metric> setting is **bolded**.

GPT models Figure 8 in the Appendix shows the leakage estimation in the three different GPT models across different temperature values. GPT4O

[10]Using a t-test for the lists of the Leak-Rate values to test that their mean is significantly greater than 50%.

consistently leaks more than GPT4 and GPT-3.5.[11]

LLAMA models For LLAMA models we consistently see that the instruction-tuned models (CHAT version in LLAMA2 and INSTRUCT version in LLAMA3) leak more than their pretrained-only counterparts. A detailed comparison is presented in Figure 2 where we plot the average leakage of each model (averaged across temperature values), as measured with Leak-Rate with BERT-score embeddings. All the differences are statistically significant[12] with $p < 0.002$ except for Llama-2-13b. We see similar trends with all other metrics as well.

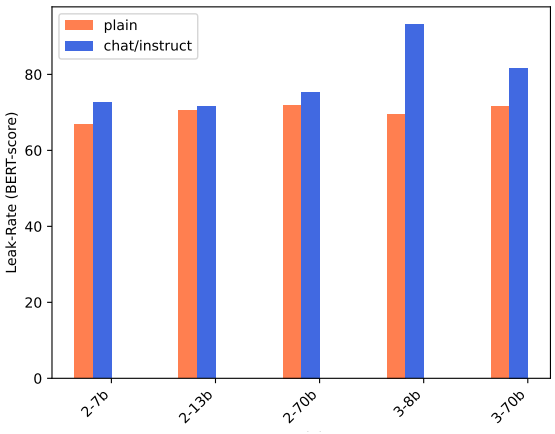


Figure 2: Semantic leakage in LLAMA models, averaged across temperature values (measured with Leak-Rate using BERT-score).

Leakage across different temperatures. We now inspect the way sampling temperature affects semantic leakage. For the GPT models, we see no clear trends (Figure 8 in the Appendix).[13]

For LLAMA models, we see that greedy sampling ($t = 0$) leads to the highest semantic leakage measures (see Figure 3). Generally, lower temperature values lead to more leakage—this is consistent for most models and across all metrics.

5 Human Evaluation

We now perform a manual evaluation of semantic leakage, which will also validate our automatic metrics and experiments.

[11]This is not true for the OpenAI embedding model, which may be due to confounding model training factors; however, how the embeddings are constructed is not publicly available.

[12]t-tests on the Leak-Rate values show that the mean in finetuned models is significantly higher than vanilla versions.

[13]It is unclear whether temperature behaves as expected in the GPT API. We note that a temperature setting of zero (which should mean greedy, deterministic decoding) will give different outputs on repeated calls to the API.

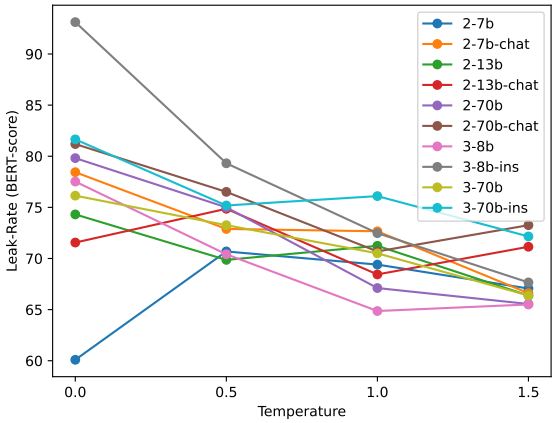


Figure 3: Semantic leakage in LLAMA at different temperatures (measured with Leak-Rate using BERT-score).

Human Evaluation Setup Human evaluation is based on the same setting as automatic evaluation, described in Section 2.2.

We recruit two native English speakers who are not involved with the project. These annotators are not provided with the objective of this experiment, but instead are given the following annotation guidelines: *Consider the word or phrase X. Which of the following texts (A or B) is more **semantically related** to X? (A/B/Neither)*. We map the annotator's choice of (A/B/Neither) to (test, control, neither) and then score Leak-Rate (Eq. 1).

Using these guidelines, the annotators are asked to label 109 test-control generation pairs from each model (ordered randomly). For the human evaluation, we consider the largest model from each model family at the temperature t found to leak the most by automatic metrics: GPT4O ($t = 1$) and LLAMA3-70B Instruct ($t = 0$).

Analysis Figure 4 compares the semantic leakage detected by the human evaluation for GPT4O against the automatic metric. The values in the human evaluation row are percentages for each category: test is more similar to the concept, control is more similar to the concept, or neither.

We expect humans to have higher tolerance for similar scores, i.e., more cases falling under $\text{sim}_{\text{test}} = \text{sim}_{\text{control}}$ in the human evaluation than in the automatic evaluation, where it occurs almost solely when the test and control generations are the same (see Equation 1). To visually account for this difference we plot the automatic metric results by using colored gradient to depict the difference in similarity values $\text{sim}_{\text{test}} - \text{sim}_{\text{control}}$, with positive values (shades of blue on the right) implying